SUKRUTH S

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PROFESSIONAL SUMMARY

AI and Data Engineer with hands-on experience in semantic and vector search, retrieval pipelines, ETL data pipelines, and applied machine learning. Experienced in developing OpenSearch-based retrieval systems using embedding models, document chunking, and vector indexing, and in building scikit-learn model-evaluation workflows. Skilled in Python, SQL, Pandas, NumPy, Java, and TypeScript, with strong foundations in REST API design, query and performance optimization, search analytics, data structures and algorithms, and CI/CD. Reduced OpenSearch query latency by nearly 50% and processed 250K+ records across CSV, JSON, and Parquet.

EXPERIENCE

Trumetric.ai | Bengaluru

Graduate Engineer Trainee (AI Search & Backend) Oct 2025 - Present

- Engineered a semantic and vector search pipeline on OpenSearch spanning document ingestion, chunking, embedding generation, and vector indexing to enable retrieval over unstructured documents.
- Optimized OpenSearch query performance, cutting latency by nearly 50% through query optimization, performance tuning, and bottleneck analysis, improving responsiveness and observability.
- Instrumented search analytics via database-backed REST APIs, capturing query latency, zero-result trends, usage metrics, and search-quality signals; introduced latency buckets for OpenSearch-specific reporting.
- Designed typo-suggestion logic based on edit distance, bounded to three character edits, to improve the search experience.
- Authored modular backend services in Java 21 and Quarkus, applying object-oriented design and service/API integration patterns.
- Integrated the search engine into Payload CMS demo sites, delivering client-facing search with TypeScript, React.js, and REST APIs.
- Contributed to real-time collaborative workflows using WebSockets, Y.js, and Redis-backed messaging, handling synchronized state and dependency validation.

XCEL Corp | Bengaluru

Full Stack Python Intern Nov 2024 - May 2025

- Developed Python and Flask backend functionality with MySQL persistence, verifying REST API requests and responses through Postman testing.
- Refactored React interfaces and backend services using maintainable, scalable patterns, reducing page-load times by 15% across key user flows.

Zidio Development | Bengaluru

Full Stack Web Development Intern Jul 2024 - Oct 2024

- Delivered a full-stack job portal on React.js, Node.js, Express, and MongoDB, integrating frontend workflows with backend REST APIs.

TECHNICAL SKILLS

Languages: Python, SQL, Java, TypeScript, JavaScript

AI / ML: Machine Learning, Regression, Model Evaluation (MAE, MSE, RMSE, R^2), Feature Engineering, Feature Preprocessing, scikit-learn, Pandas, NumPy, Embeddings, Embedding Models, RAG, Semantic Search

Search & Retrieval: OpenSearch, Vector Search, Vector Indexing, Vector Databases, Document Chunking, Document Ingestion, Query Optimization, Information Retrieval, Search Analytics

Data Engineering: ETL, Data Pipelines, Data Ingestion, Data Cleaning, Data Transformation, Data Aggregation, Parquet, CSV, JSON, Producer-Consumer Queues, Analytics Workflows

Databases: MySQL, PostgreSQL, MongoDB, Redis, SQLite, SQL Aggregation

Backend & APIs: Flask, Django, Quarkus, Node.js, Express.js, REST APIs, API Design, Webhooks, WebSockets, Y.js, Authentication; Frontend: React.js, HTML5, CSS3, Payload CMS

Tools & DevOps: Git, GitHub, GitHub Actions, CI/CD, Postman, API Testing, Debugging, Performance Tuning

Core: Data Structures & Algorithms, Object-Oriented Programming, Modular Design, HTTP/HTTPS, Real-Time Systems

PROJECTS

- **Pharma Intelligence & Data Analytics Pipeline** - Architected an ETL pipeline ingesting 250K+ medicine records from CSV, JSON, and Parquet using Python, Pandas, NumPy, and MySQL, with format-specific ingestion and reusable transformation components. Decoupled processing from database writes via producer-consumer queues and precomputed aggregates by medicine, category, state, sales, and usage to serve analytics without rescanning raw data.
- **Student Performance Analysis using Machine Learning** - Designed a supervised regression workflow in Python and scikit-learn to predict math scores, using OneHotEncoder and StandardScaler for feature preprocessing. Benchmarked six algorithms on MAE, MSE, RMSE, and R-squared, and selected Linear Regression as the final model based on R-squared performance.
- **ProMentor Platform** - Created a full-stack mentorship platform with Flask REST APIs, MySQL, and React.js, featuring token-based authentication, mentor listings, and session booking across a layered frontend, API, and database architecture.

EDUCATION

Bachelor of Engineering in Computer Science & Engineering | 2021 - 2025 | CGPA: 8.53

CERTIFICATIONS

- Microsoft Learn - Introduction to GitHub Actions
- HackerRank Python (Basic) Certification
- Google Cloud Arcade - hands-on labs and challenges using GCP services